

Predicting new COVID-19 cases in the US: A Comparison of a Compartmental model (SIR) and XGBOOST Machine Learning Model

Basil Okola

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Background: Figuring out the number of new COVID-19 cases was key in aligning interventions to hotspots. This included ensuring there were enough beds for admissions, key protection gear like face masks, oxygen supply and intensive care unit space. A lot of mechanistic models were build borrowing from the traditional SIR model. Additionally, researchers explored machine learning techniques to predict the course of the infections.

Methods: In this report, we compare the performance of a SIR model to an XGBOOST prediction model in predicting new cases in each of the American states, 3 years after the pandemic.

Results: The XGBOOST ML approach demonstrated robust goodness-of-fit, effectively capturing the non-linear peak dynamics across varying geographic replicates.

1 INTRODUCTION

We chose to use data on COVID-19 cases assembled by a team from [The London School of Health & Tropical Medicine](#) who built a global COVID-19 tracker shiny app. I was already working on a personal project that modularized parts of the said app into a golem framework and so it was easier adding a prediction modelling capabilities rather than starting a new project. The [COVID19Dash](#) app can be cloned from github and run seamlessly. For predictions look at the `Prediction Model` tab.

2 METHODS

2.1 Data

We used `cv_states` data that is part of the datasets in the `COVID19Dash` package. It compiles daily COVID-19 cases in each of the states in the US. It also comes with limited spatial support (longitudes & latitudes). We chose `new_cases` as the outcome variable and generated rolling means for 7, 14, and 30 days as predictors in addition to the frequency of each state in the data to avoid having to encode the categorical variable.

2.2 Models

We fitted a SIR model - a compartmental mechanistic framework that simulates the spread of infectious diseases by transitioning a population through susceptible, infectious, and recovered states using differential equations. It relies on the interaction between the transmission rate β and the recovery rate γ to determine the velocity and peak of an epidemic curve¹.

2.2.1 mathematical framework

The mechanistic sir model assumes a fixed population N where individuals transition between susceptible (S), infectious (I), and recovered (R) compartments. the dynamics are governed by the following system of ordinary differential equations, which were implemented using the `desolve` package in our workflow:

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

where β represents the effective transmission rate and γ denotes the removal or recovery rate. individuals are assumed to move from the susceptible to the infectious compartment at a rate proportional to the product of S and I .

2.2.2 basic reproduction number

A critical threshold in this modeling framework is the basic reproduction number (R_0), which defines the average number of secondary infections produced by a single infected individual in a completely susceptible population. in our analysis, we calculated R_0 using the optimized parameters from each state:

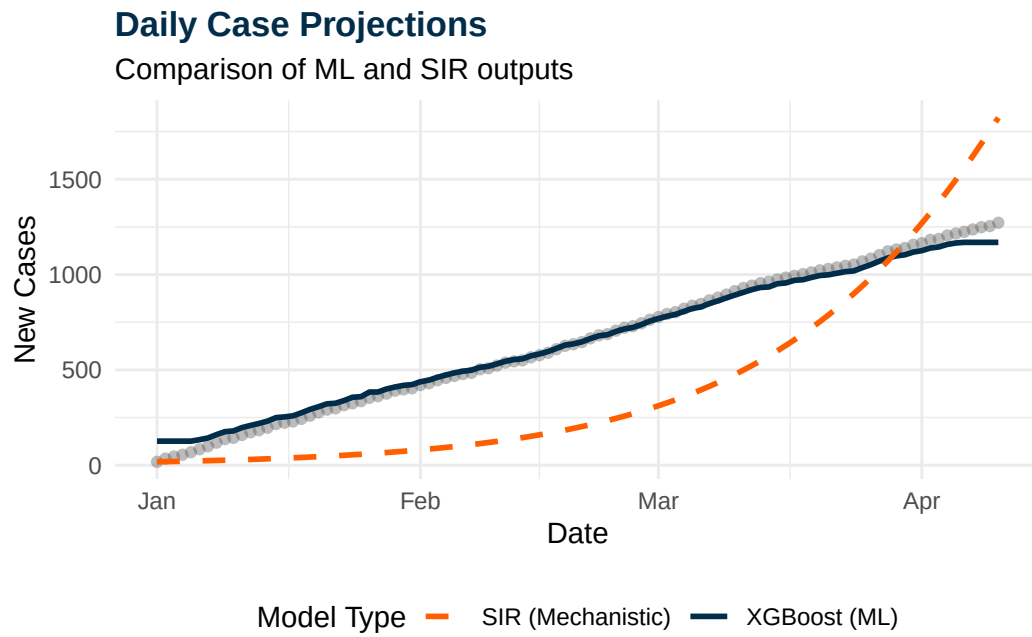
$$R_0 = \frac{\beta}{\gamma}$$

An epidemic persists only when $R_0 > 1$. our model fitting used an optimization routine to minimize the residuals between the predicted $I(t)$ compartment and the observed case ratios across the 50 american states.

2.3 Missing data

2.4 Accounting for spatial and temporal structure

Model comparisons



Summary of parameters

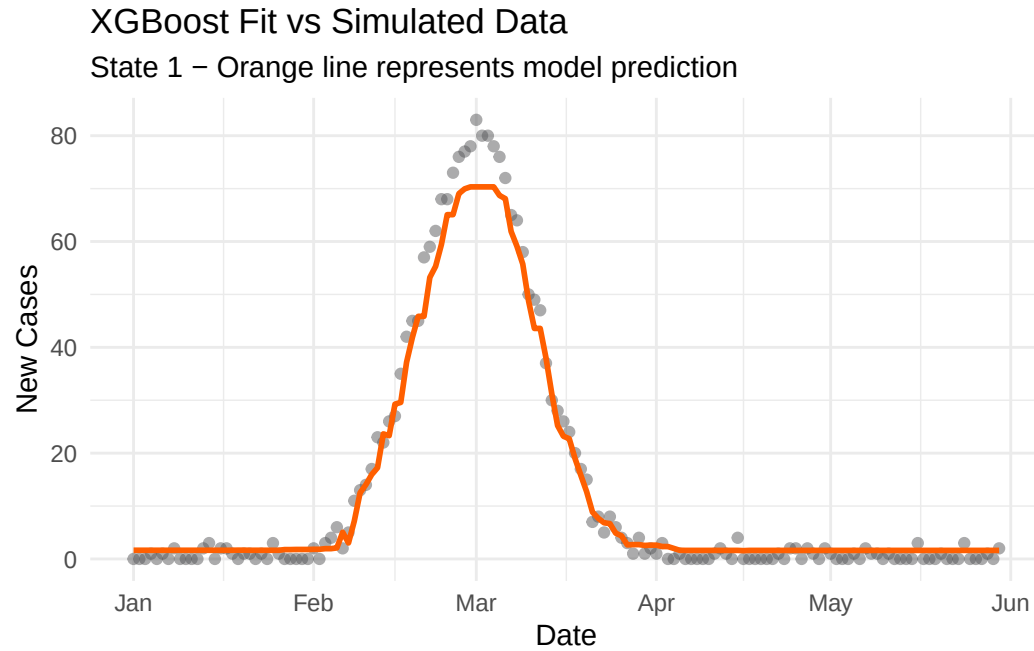
Table 1: SIR Model Parameters

Parameter	Value
Beta (Transmission)	0.5159
Gamma (Recovery)	0.4663
Basic Reproduction (R_0)	1.1065

Simulations

Table 2: Global Assessment Metrics

rmse	mape
1.507113	112.8102



- 1 Ross R. An application of the theory of probabilities to the study of a priori pathometry.—part i. *Proceedings of the Royal Society of London Series A, Containing papers of a mathematical and physical character* 1916; **92**: 204–30.